

Apollo F-1 Engine Tacit Knowledge Loss — Sources

This document lists the primary and secondary sources cited in the case study.

Primary Sources — NASA Documentation

- NASA Marshall Space Flight Center (2013). F-1 Engine Scanning and Modeling Project. Huntsville, AL.
 - Technical report on 3D laser scanning of museum F-1 engines and analysis of archived documentation.
 - NASA (1969). Saturn V Flight Manual. NASA Report SA 503.
 - Original operational documentation for Saturn V and F-1 engines.
 - Rocketdyne (1967). F-1 Engine Familiarization Training Manual. Canoga Park, CA: Rocketdyne Division, North American Aviation.
 - Technical training materials for Apollo-era engineers and technicians.
-

Secondary Sources — F-1 Engine History and Technical Analysis

- Young, Anthony (2009). The Saturn V F-1 Engine: Powering Apollo into History. Chichester, UK: Springer Praxis.
 - Comprehensive technical and historical account of F-1 development, manufacturing, and legacy.
 - Bilstein, Roger E. (1980). Stages to Saturn: A Technological History of the Apollo/Saturn Launch Vehicles. NASA SP-4206. Washington, DC: NASA History Office.
 - Official NASA history of Saturn V development, including F-1 engineering challenges.
 - Schoenman, Edward M. (2015). "The F-1 Engine: A Design Classic." Air & Space Smithsonian, July 2015.
 - Popular account of F-1 engineering and its enduring influence.
-

Tacit Knowledge and Manufacturing Capability

- Bergin, Chris (2013). "NASA Resurrects F-1 Engine Design with Modern Tech." NASASpaceFlight.com, April 16, 2013.
- Report on NASA's 2013 scanning project and conclusions about restart feasibility.

- NASA Marshall Space Flight Center engineers (2013). Interviews and presentations on F-1 heritage and modern applications.
- Cited in NASA 2013 report and contemporary aerospace journalism.
- Loff, Sarah (2013). "NASA Tests Vintage Engine, Eyes New Rocket Booster." NASA.gov, August 28, 2013.
- Report on gas-generator test using recovered F-1 components.

Tacit Knowledge Theory

- Polanyi, Michael (1966). The Tacit Dimension. University of Chicago Press.
- Foundational work on tacit versus explicit knowledge; "we know more than we can tell."
- Nonaka, Ikujiro & Takeuchi, Hirotaka (1995). The Knowledge-Creating Company. Oxford University Press.
- Framework for organizational knowledge creation and transfer, including tacit knowledge capture.
- Collins, Harry (2010). Tacit and Explicit Knowledge. University of Chicago Press.
- Philosophical and sociological analysis of tacit knowledge transmission and loss.

Aerospace Engineering and Legacy Systems

- Launius, Roger D. & Jenkins, Dennis R. (2012). Coming Home: Reentry and Recovery from Space. NASA SP-2011-593.
- Broader context of Apollo-era engineering and post-Apollo capability loss.
- McCurdy, Howard E. (1993). Inside NASA: High Technology and Organizational Change in the U.S. Space Program. Johns Hopkins University Press.
- Institutional analysis of NASA knowledge retention and organizational memory.
- Augustine, Norman R., et al. (2009). Seeking a Human Spaceflight Program Worthy of a Great Nation. Washington, DC: NASA Human Spaceflight Plans Committee.
- Report addressing loss of Apollo-era capabilities and recommendations for future programs.

Modern Rocket Engineering (Comparative Context)

- SpaceX (2020). "Merlin Engine." Technical specifications and development history.
- Modern rocket engine design philosophy; different approach from F-1.
- Aerojet Rocketdyne (2015). "RS-25 Engine." Technical documentation.
- Space Shuttle main engine, derived from Apollo-era technology, adapted for SLS.

- Sutton, George P. & Biblarz, Oscar (2016). Rocket Propulsion Elements, 9th ed. Wiley.
 - Standard aerospace engineering textbook; includes historical and modern engine design.
-

CIF and Knowledge Inheritance Context

- Vincenti, Walter G. (1990). What Engineers Know and How They Know It. Johns Hopkins University Press.
 - Epistemology of engineering knowledge; relevance to tacit/explicit knowledge distinction.
 - Ferguson, Eugene S. (1992). Engineering and the Mind's Eye. MIT Press.
 - Role of non-verbal, visual, and tacit knowledge in engineering practice.
-

Last updated: 2026-07-28

Series: CIF Recovery Systems Test Series

Author: Derek Hone · Remnant Fieldworks Inc.